BIOCHEMISTRY CUMULATIVE EXAMINATION

November 10, 1990

Hydropathy plots are often used to predict membrane spanning regions of transmembrane proteins. The use of these plots assumes that the transmembrane regions are alpha helical in nature.

Consider the following (20 points each):

a. What is a hydropathy plot and how is it interpreted?

b. Describe two features of an alpha helical structure which would make it an energetically favorable structure capable of spanning a biological membrane.

c. Describe at least two experimental observations supporting alpha helical structures as membrane spanning regions in transmembrane proteins.

d. Some transmembrane helical regions are amphipathic in that they have one side which is predominantly hydrophobic while the other side is polar in nature. How might such helical structures be used to span the membrane? Would a hydropathy plot predict their existence? Why or why not?

e. Despite the widespread use of alpha helical segments as transmembrane regions, some proteins span the membrane using other structural motifs such as beta sheet structures. Explain how the structural characteristics of the beta sheet would allow it to also span a biological membrane. How might a amphipathic beta sheet structure be accomodated in a biological membrane?